

ONLINE JOB FAIR APPLICATION

Project Report

DECLARATION

I hereby declare that the project work entitled "**Online Job Fair Application**" is an original effort carried out by our team during the Hackathon event. The work presented here is genuine and has not been submitted to any other competition, university, or institution for any academic or professional credit. This project was developed solely for learning purposes, to explore full-stack development using real-world technology tools including **Spring Boot, PostgreSQL, Spring JPA, and Thymeleaf**.

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ABSTRACT

The **Online Job Fair Application** is a web-based platform designed to digitalize the traditional job recruitment fair experience. In conventional job fairs, job seekers must physically visit venues, submit printed resumes, and wait for multiple interview rounds, leading to time consumption, travel costs, and inefficiency. Our system eliminates these challenges by offering a centralized online portal where **companies can post job openings and review applications**. Similarly, job seekers can **create profiles, upload resumes, search job listings, and apply seamlessly**.

The system follows the **Model-View-Controller (MVC)** architecture using **Spring Boot (backend)** and **Thymeleaf (UI templating)** integrated with **PostgreSQL as a secure relational database**. We implemented role-based login for companies and applicants, real-time application status tracking, filtering and searching of job opportunities, and resume upload capability. This project demonstrates hands-on experience in building a real-world full-stack application suitable for employment platforms, recruitment drives, and campus placement systems.

INTRODUCTION

With digital transformation accelerating across industries, the recruitment sector has also shifted from manual job fairs to online hiring platforms. Traditional job fair processes involve large crowds, manual resume collection, long queues, and delays in result announcements. These inefficiencies motivated us to design a **digital job fair ecosystem** that connects job seekers and recruiters virtually. The system ensures easy job publishing by organizations and flexible job search options for candidates from anywhere without physical participation.

Our Online Job Fair Application provides a **simple, secure, and intuitive user interface**, making it convenient for both technical and non-technical users. It supports **organized resume handling, reduces administrative effort, increases reach for rural and remote candidates, and promotes transparency in hiring status**. This platform reflects real corporate hiring systems and also demonstrates our understanding of software engineering concepts, full-stack development, databases, and deployment workflows.

OBJECTIVES OF THE PROJECT

The main objective of this project is to build a **digital recruitment management system** that connects job seekers with employers efficiently. The key goals are:

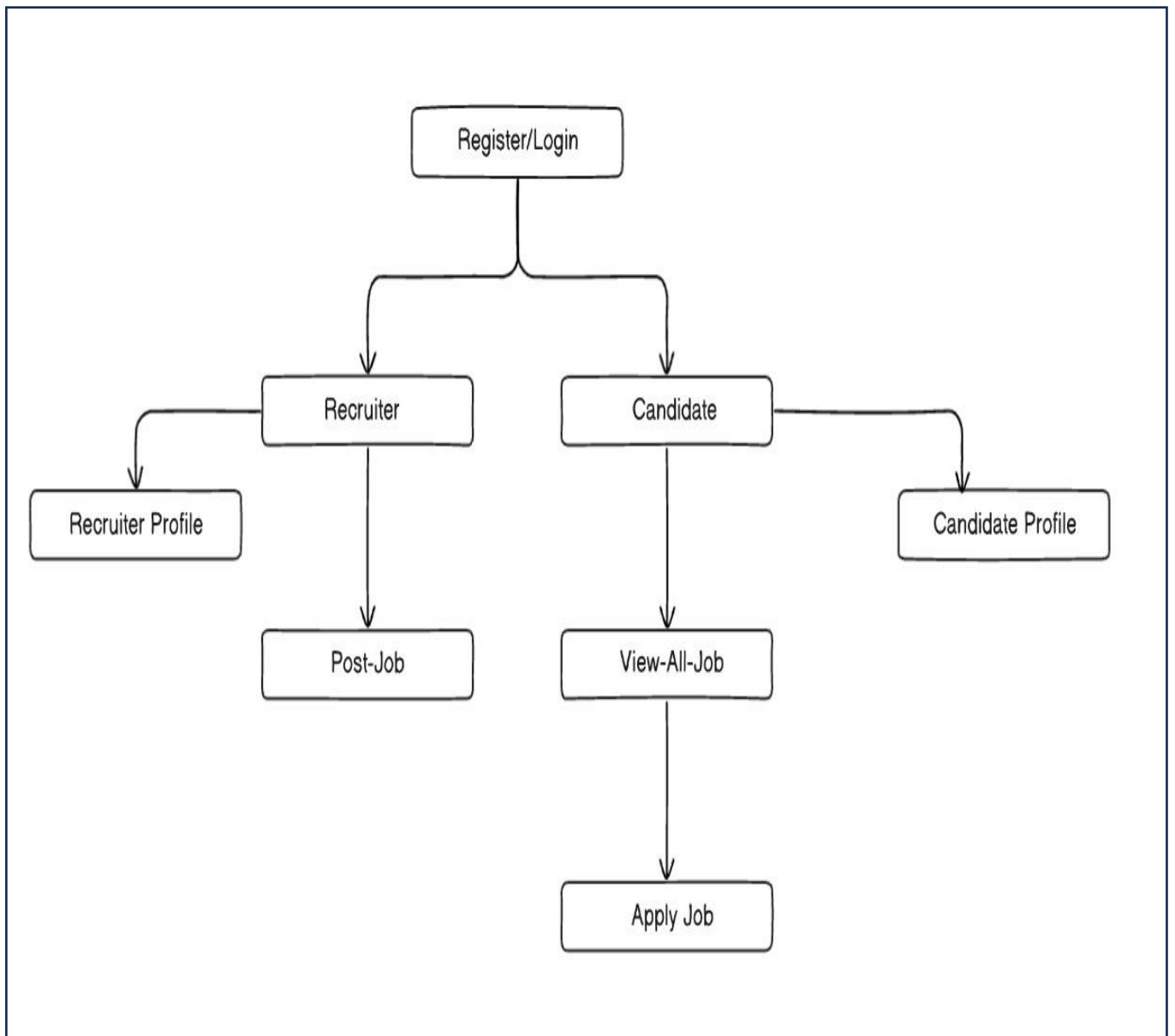
- To provide an **accessible and time-efficient online job fair experience** for candidates and companies.
- To implement **secure login and role management** for job seekers and recruiters.
- To allow companies to create professional profiles and **post job openings with skill requirements, eligibility criteria, and role details**.
- To enable job seekers to **create digital profiles, upload resumes, and apply for positions** based on interest and qualifications.
- To digitalize the application process with **real-time status updates (Applied, Shortlisted, Rejected, Selected)**.
- To ensure easy data storage and retrieval using **PostgreSQL database with relational tables**.
- To practice full-stack development principles and integrate backend, database, and UI modules effectively.

Hardware & Software Requirements

- This project is designed to run efficiently on commonly available hardware and open-source tools, making it accessible for academic and enterprise environments.

Category	Specifications
Frontend	Thymeleaf, Bootstrap, HTML, CSS, JS
Backend	Java Spring Boot Framework
Database	PostgreSQL
API Layer	Spring JPA
IDE Used	IntelliJ IDEA / VS Code
Java Version	Java JDK 17
Minimum Hardware	4GB RAM, 1.5 GHz processor
Operating System	Windows

WORK FLOW DIAGRAM



MODULES

1. Authentication & Role-Based Authorization Module

This module is responsible for validating users and managing access privileges. It separates **Company accounts and Job Seeker accounts** to ensure secure and role-specific functionality. Passwords are processed securely and login permissions are controlled using Spring Security principles.

2. Company Job Posting Module

This module allows companies to create and post job vacancies on the platform. Companies can enter important job details such as job title, description, required skills, location, and salary package. After posting, the job becomes visible to job seekers.

Only verified company accounts can post jobs, ensuring genuine listings. Companies can also edit or delete their posted jobs anytime. This module helps organizations reach suitable candidates easily and manage hiring requirements efficiently.

3. View All Job Module

This module displays all active job openings in one place for job seekers. It shows key information like job title, company name, skills needed, and location in a clean and organized format.

Job seekers can quickly view all available jobs, check details, and apply directly from this page. This module helps candidates explore opportunities easily and speeds up the job search process.

4. Apply Job Module

Candidates can apply to jobs directly from the portal. The system records job details, applicant information, and time stamp in the database, preventing repeated applications for the same position.

5.Candidate Profile Module

This module allows job seekers to create and manage their profile. Candidates can add their personal details, skills, education, experience, and upload their resume. They can update their information anytime. This profile helps recruiters quickly understand the candidate's background before reviewing applications.

6. Recruiters Profile Module

This module lets companies create and update their company profile. Recruiters can add company name, description, location, contact details, and logo. This makes the platform trustworthy and helps candidates see verified company information before applying for jobs.

CONCLUSION

The Online Job Fair Application successfully demonstrates the digital transformation of the traditional job fair process. It allows seamless interaction between job seekers and recruiters through a structured online system. The platform reduces manual paperwork, improves communication, and supports remote hiring, which is essential in modern recruitment landscapes.

This hackathon project strengthened our understanding of **end-to-end web development, MVC architecture, database management, REST integration, and teamwork in agile conditions**. With enhancements like automated resume scoring, chatbot support, and online scheduling, it has potential to evolve into a full professional job marketplace system.